

Akash Gogate

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EDUCATION

University of Wisconsin at Madison

Graduating: May 2028

Bachelor of Science in Computer Science and Biology • Dean's List Fall 2024, Spring 2025, Fall 2025 • 3.9 GPA

Coursework: Artificial Intelligence, Deep Learning, Advanced Algorithms, Linear Algebra, Differential Equations, Data Structures, Cryptography, Computer Science Capstone, AP Statistics, AP Biology, AP Computer Science A

TECHNOLOGIES

Languages: Python, R, Java, JavaScript, Rust, C++, SQL, Unix/Linux/Bash

ML & Statistics: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Hugging Face, NLP, Neural Networks, Random Forest, KNN, Biostatistics, Multi-Omics Data Integration, Statistical Modeling, Statistical Analysis, Cross-Validation, Statistical Significance, Data Analysis, Data Visualization

Bioinformatics: Scanpy, Seurat, CARD, Bioconductor, DESeq2, scVI, scANVI, scGen, AmortizedLDA, scRNA-seq, Single-Cell RNA Sequencing, Single-Cell Analysis, NGS, Next-Generation Sequencing, Spatial Transcriptomics, Computational Biology, Genomics, Gene Expression, Differential Expression, Pathway Analysis, Pipeline Development, Reproducible Research, Pipeline Validation

Dev Tools: Git, GitHub, GitHub Actions, Version Control, Jupyter Notebook, FastAPI, Weights & Biases, LLM APIs, RAG

RESEARCH EXPERIENCE

The Kendzierski Lab at the University of Wisconsin-Madison

September 2025 - Present

Student Research Intern

- Built hypothesis-driven agentic pipeline for single-cell RNA sequencing (scRNA-seq), next-generation sequencing (NGS), and spatial transcriptomics genomics analysis — including gene expression and differential expression — across scVI/scANVI/scGen/AmortizedLDA on Linux/Unix; GPU-accelerated profiling cut consumption by **300%**
- Engineered TransferAgent for interdisciplinary paper grading: automated citation mapping, literature synthesis, cross-paper validity scoring, data analysis, and version control; reproducible evaluation methodology reduced review time by **~60%**
- Collaborated with clinicians to validate pipeline efficiency gains; generated data visualizations quantifying scRNA-seq computational biology improvements across drug discovery and precision medicine workflows

Inspirit AI Research Internship

Sep 2022 - Mar 2023

Machine Learning Research Intern

- Built skin cancer detection pipeline (OpenCV + scikit-learn/TensorFlow, NumPy/Pandas) achieving **93% classification accuracy**; published as peer-reviewed paper through Inspirit AI
- Trained Random Forest on miRNA sequences achieving **95% predictive accuracy ($p < 0.05$)**; validated via stratified k-fold cross-validation

PUBLICATIONS

"Early Skin Cancer Detection Improvement," Inspirit AI Research Program, 2023

TECHNICAL PROJECTS

Self-Improving LLM Agent (<https://github.com/AkashGogate/SelfImprovingLLMAgent>)

Jan 2026 - May 2026

- Developed novel **repeated failure rate** metric across **5,000+ task benchmark**; LLM-as-judge harness scored 7 quality dimensions; outperformed ReAct by **15pp**

Lotus Health (<https://github.com/AkashGogate/lotus-health>)

March 2026

- Built disease cost simulation engine on **1,080-node ICD-10 comorbidity graph** across **45M MEPS/CMS records** via SQL; RAG pipeline maps patient symptoms to disease risk profiles via data analysis
- Hybrid architecture separates NLP inference from deterministic logic; outputs are **ACID-compliant** and hallucination-free — critical for clinical data integrity

miRcore Cancer Detection Research

Jul 2021

- Trained and validated Random Forest in R on micro-RNA expression sequences achieving **95% predictive accuracy ($p < 0.05$)**; validated via stratified k-fold cross-validation

INDUSTRY EXPERIENCE

Leidos

May 2025 - Present

Software Engineer Intern

- Engineered reproducible Python scheduling optimizer via dynamic programming + memoization, **reducing constraint violations** across multi-objective real-time operational constraints
- Built fault-tolerant Kafka data pipeline to MongoDB; deployed Docker-containerized test suite on Kubernetes via **GitHub Actions CI/CD** with full unit coverage

LEADERSHIP

Princeton Racket Club

May 2024 - Aug 2024

Tennis Coach and Tournament Director

- Directed tournament logistics and scheduling; coached athletes of all skill levels with customized training programs; managed end-to-end client communications and conflict resolution throughout the season

Tennis Racket Stringing Services

Jan 2019 - Present

- Founded and scaled to **50+ clients**; manage all operations, inventory, and client consultations